

Recreating Cessna

Project Report

Wing Parameters, Fuselage Length, and Tail Sizing Method

1. Wing Parameters Used

The wing geometry for the recreated **Cessna 210 (NASA Variant)** model was defined using the basic aerodynamic parameters available in OpenVSP. These parameters control the size, shape, and aerodynamic behavior of the wing.

The **wing span** was taken from the front-view drawing and is approximately **36.748 ft**, representing the total distance from one wingtip to the other.

The **root chord** was selected based on the width of the wing where it connects to the fuselage, while the **tip chord** was reduced to create a tapered wing. This taper improves aerodynamic efficiency and reduces induced drag. The **taper ratio** was adjusted accordingly to maintain the realistic geometry of the aircraft.

A small **sweep angle** was applied to the wing to match the slight rearward sweep visible in the top view of the aircraft. This sweep helps improve stability and aerodynamic performance at higher speeds.

The **dihedral angle** was also included in the wing design. The dihedral causes the wings to tilt slightly upward from the fuselage, which improves the aircraft's **lateral stability** during flight.

Typical parameters used in the OpenVSP model include:

- Wing span: ~36.7 ft
- Root chord: ~5.5–6 ft
- Tip chord: ~2.5–3 ft
- Taper ratio: ~0.45–0.5
- Dihedral angle: ~5°
- Sweep angle: small positive sweep

These parameters allow the wing to closely resemble the real aircraft geometry shown in the reference drawings.

2. Fuselage Length

The **fuselage length** of the aircraft was obtained from the side-view drawing provided in the project document. The total fuselage length is approximately **24.580 ft**, measured from the nose to the tail end.

In OpenVSP, the fuselage was created using the **Fuselage geometry tool**, and the shape was adjusted using the **XSec Editor**. Multiple cross-sections were modified to match the aircraft profile seen in the side and top views.

The fuselage design included three main sections:

1. **Nose Section** – streamlined and tapered to house the propeller and engine.
2. **Cabin Section** – the widest portion of the fuselage, designed to accommodate passengers and cockpit.

3. **Tail Cone Section** – gradually tapering section connecting the fuselage to the tail surfaces.

The cross-sections were positioned along the **X-axis** to maintain the correct proportions while adjusting **width, height, and curvature** to match the provided drawings.

3. Tail Sizing Method

The horizontal and vertical tails were designed using a **scaled-down wing geometry approach**, which is a common conceptual design method used in aircraft modeling.

First, a new wing geometry was added and resized to create the **horizontal tail**. The span and chord were reduced compared to the main wing, and the tail was placed near the rear of the fuselage. The horizontal tail provides **pitch stability and control**.

Next, another wing geometry was created and **rotated 90°** to form the **vertical tail (fin)**. The height and sweep were adjusted to match the vertical stabilizer shape shown in the reference drawings. The vertical tail provides **yaw stability and directional control**.

The tail sizes were selected based on visual scaling from the **top and side views** provided in the dimensioned aircraft drawings. The horizontal tail span is approximately **12.99 ft**, as shown in the top-view diagram.

This method ensures that the tail surfaces maintain proper proportions relative to the fuselage and wing, allowing the model to closely resemble the real Cessna 210 configuration.

The above sections provide the theoretical dimensions approximately for the given aircraft. Some of the dimensions were obtained from image tracing as instructed. Surface smoothness and geometric proportions were prioritised.

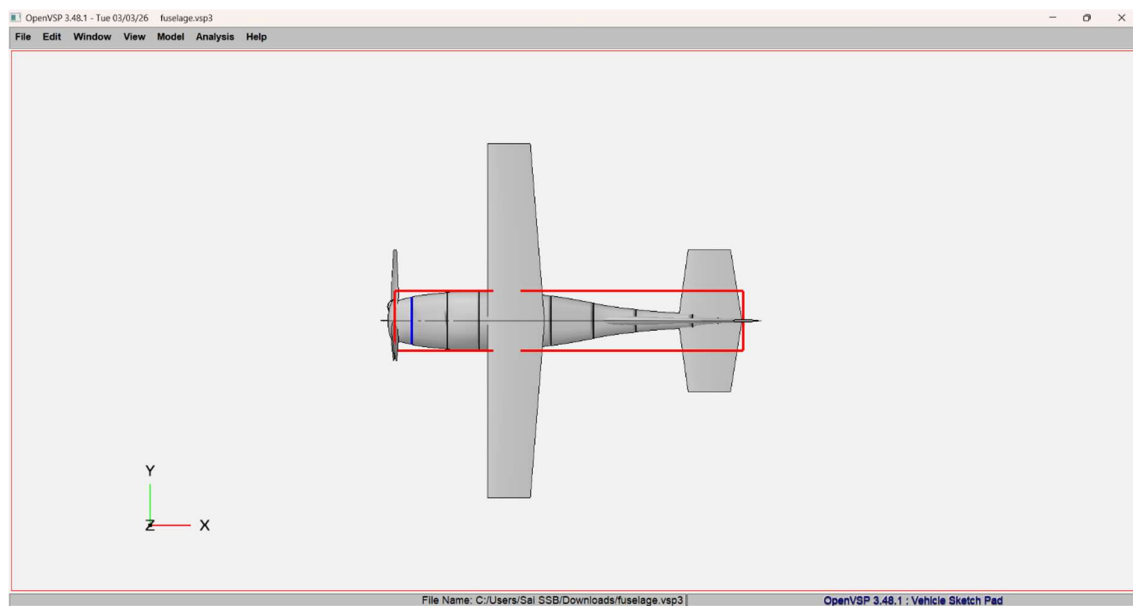
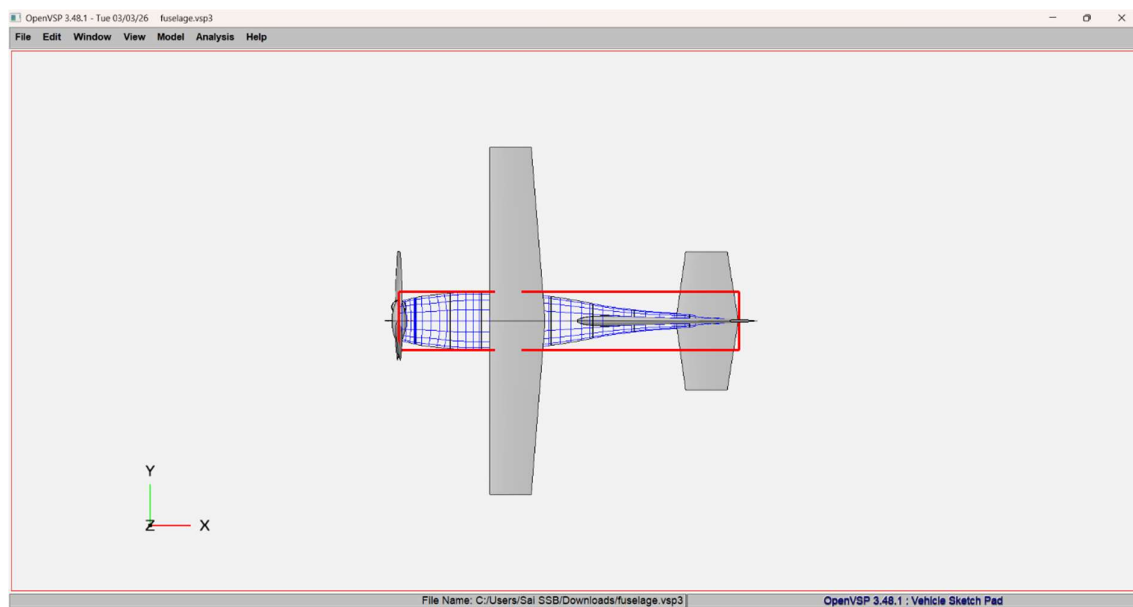


Figure 1: Top view

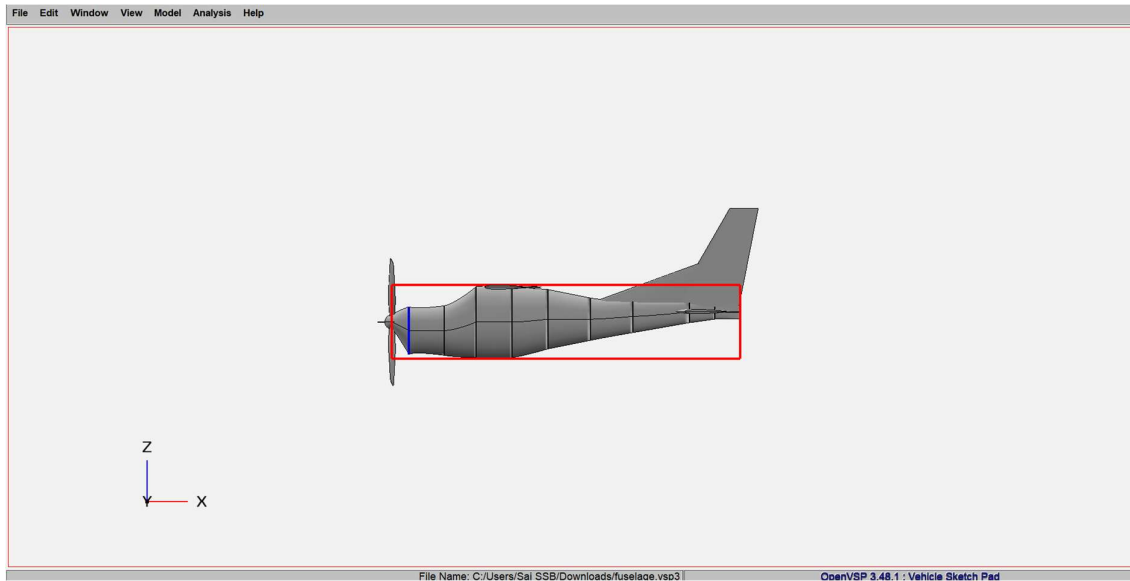
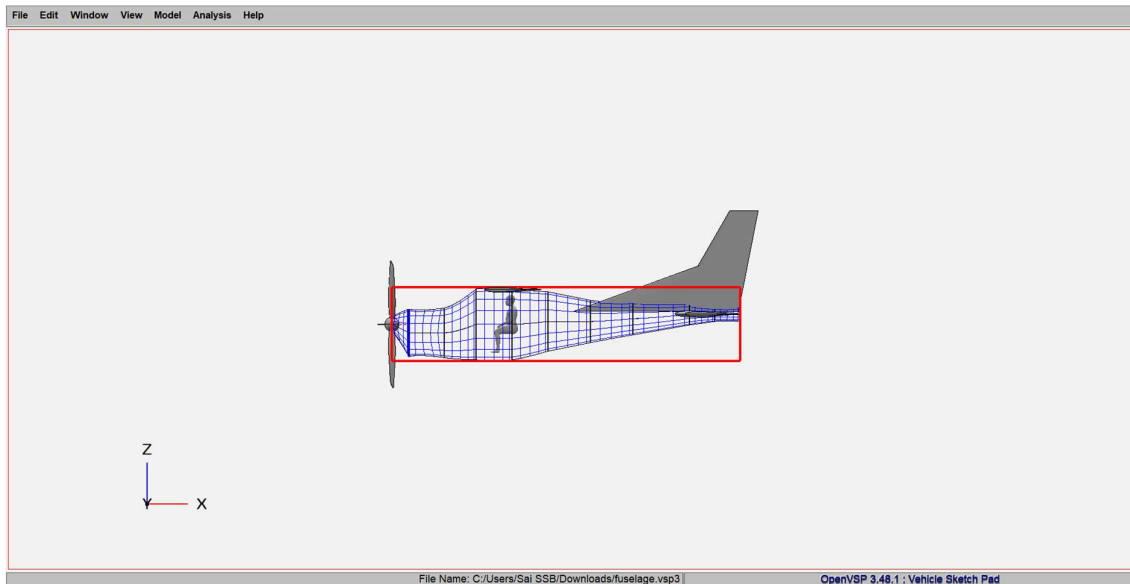


Figure 2: Side view

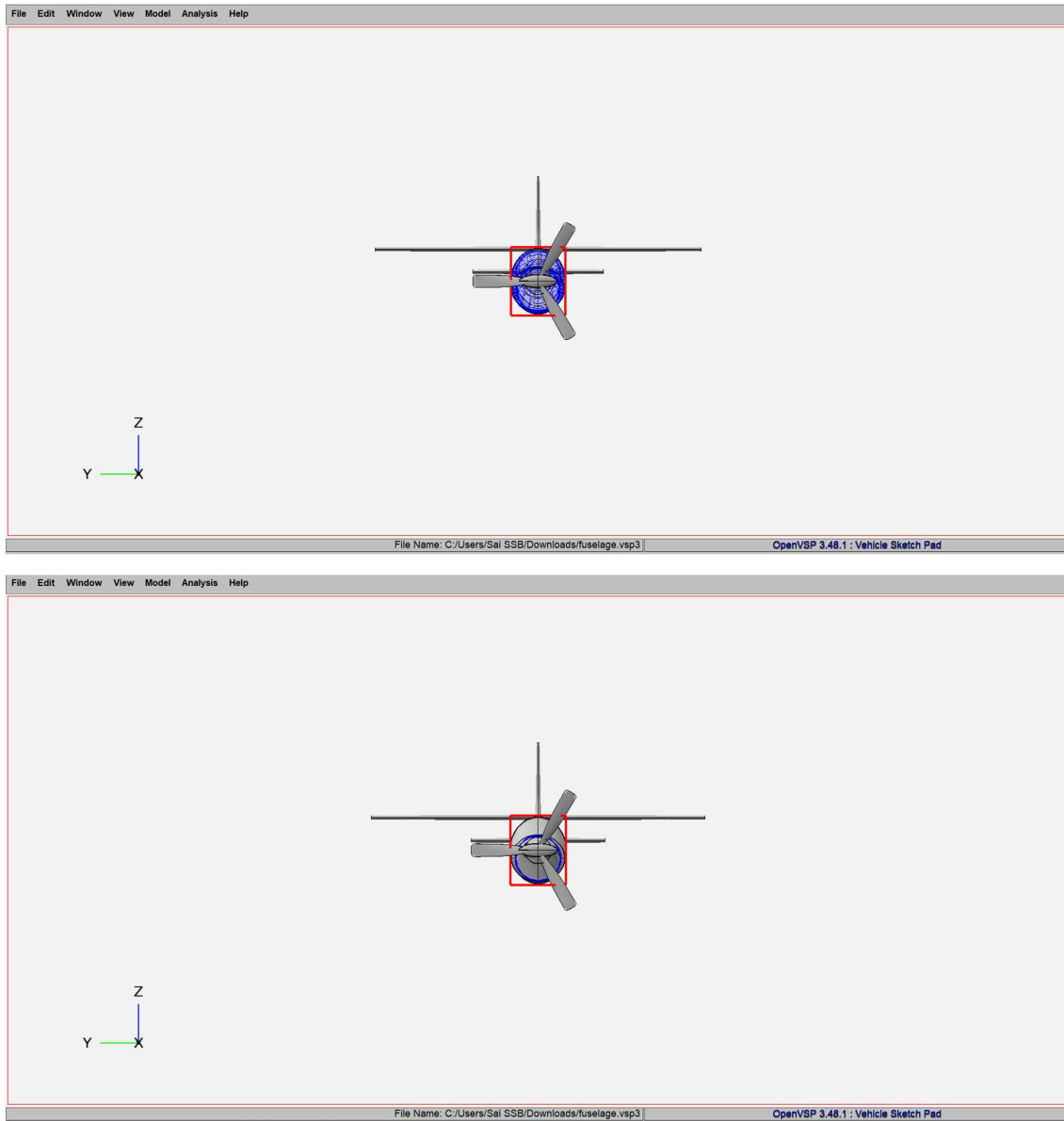


Figure 3: Front view

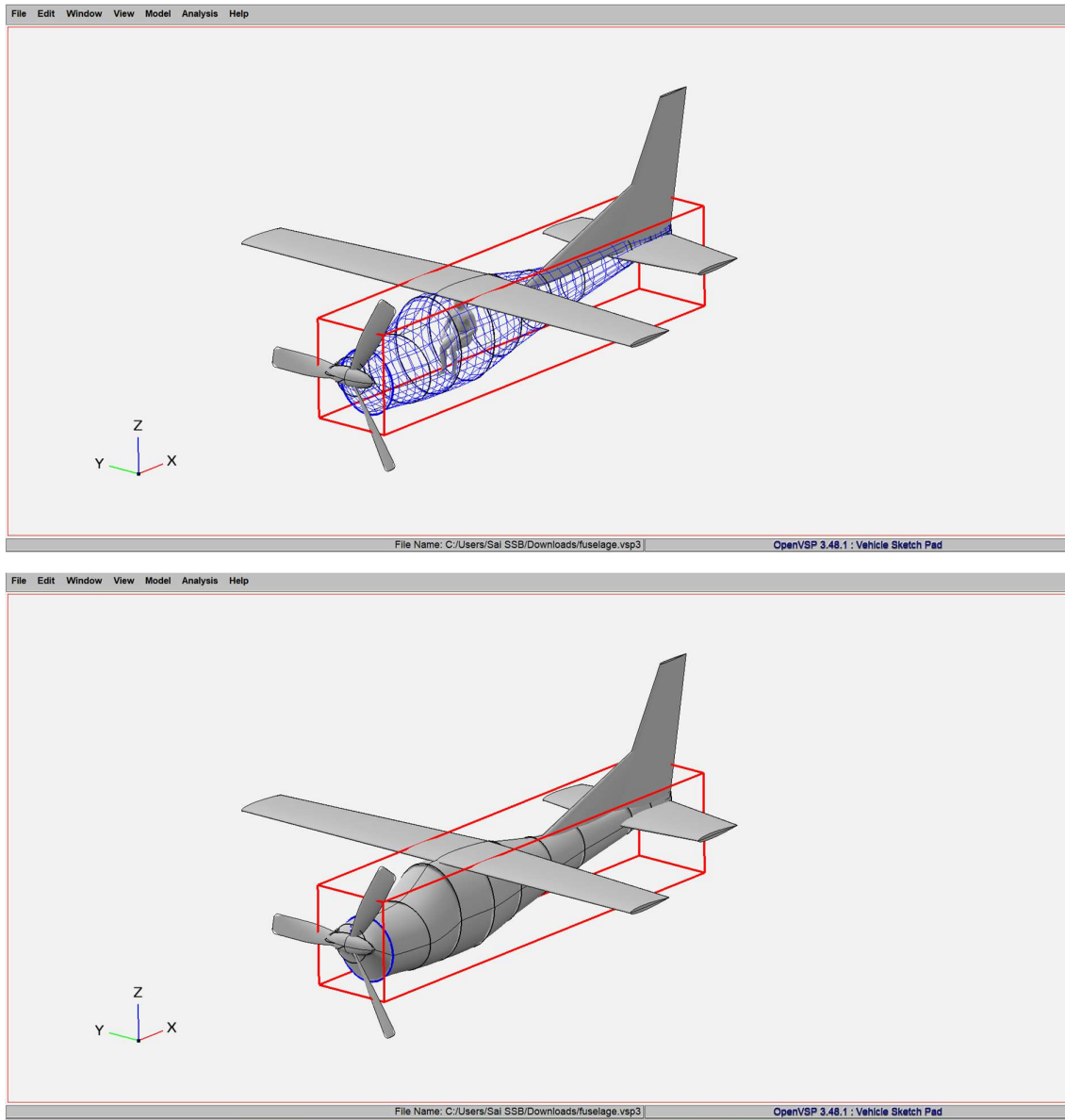


Figure 4: Aircraft Model